

UNIT - 3

Role of structural engineer in built environment :

A structural engineer also plays a major role as a team among other professionals like surveyor, quantity surveyor, and architects engineers.

The following tasks must be performed by a structural engineer:

- . Design models of structures using software.
- . Assessing the reaction of structures to pressures and stress.
- . Finalizing the appropriate concrete materials that would be suitable for the structure.

- . Assessing budget of the project.
- . To ensure that newly erected buildings are structurally sound with construction contractors.
- . Using computers and computer-aided design technology for simulation purposes.

Stability and Strength Aspect of buildings :

A structure must be both strong and stable. If it is not strong it will break ,crack or collapse. This can happen when the material is not strong enough to take the stress coming on it.

If a structure is not stable it will move or fall or slide; even if the material of the component parts has not broken (or failed).

The body of an unstable structure may remain intact without any yielding, crushing, cracking etc. But it may move as a whole (rigid body displacement). This movement can be one of the following

- Sliding
- Overturning
- Sinking

Strength is a material property. If the stress in the structure exceeds the limiting stress at any part of the structure, this condition is a state of failure.

Buckling of a compression member in a structure is also a form of failure. Theoretically this is classified as instability. In most cases of buckling (of a compression member) a member suddenly switches from compression mode to bending mode.

Earthquake resistant design aspect

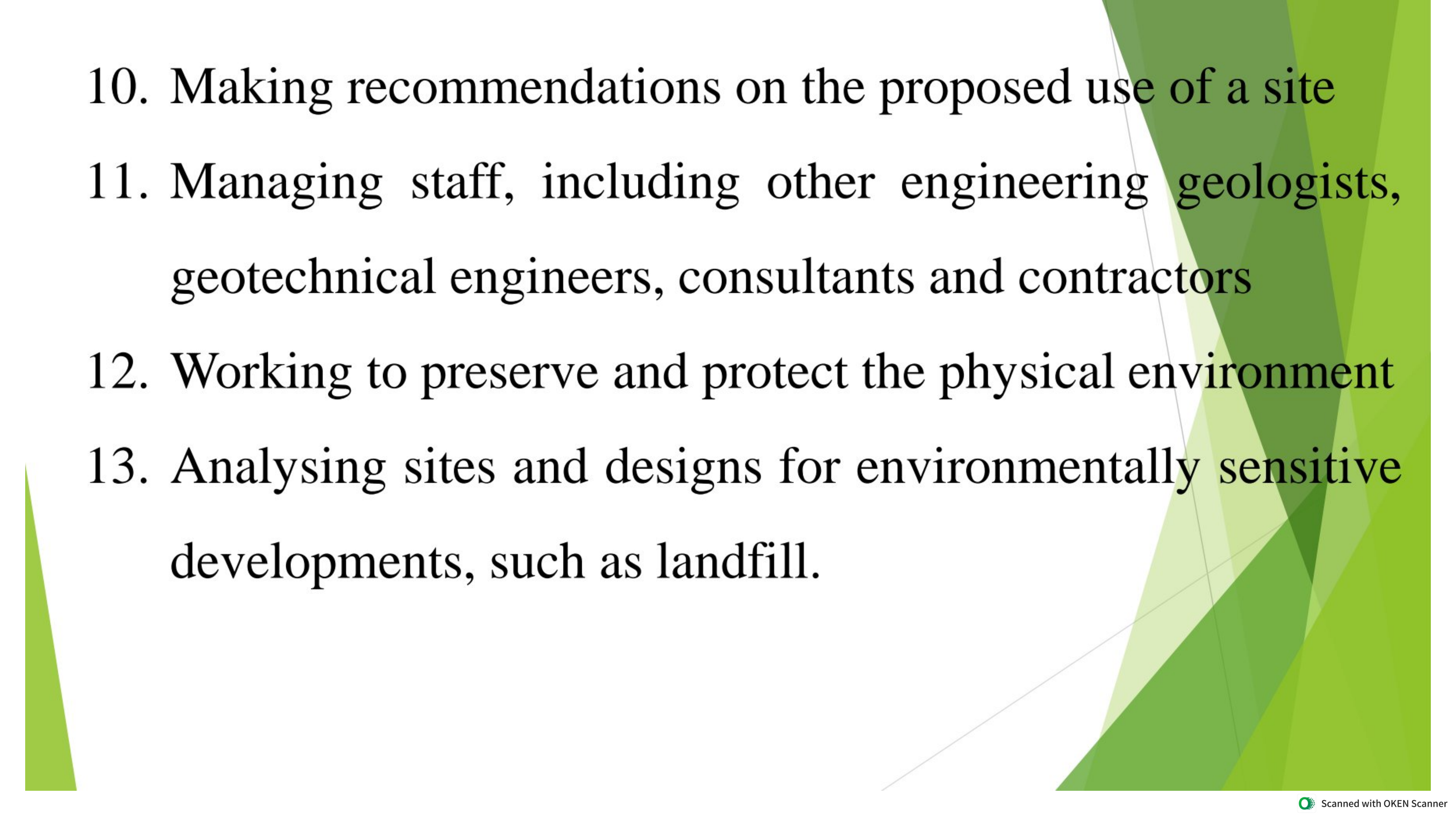
1. The building plan should be in a regular shape such as square or rectangular.
2. No wall in a room should exceed 6.0m in length. Use pilasters for longer walls. In hilly terrain, it should not exceed 3.5m in length.
3. The height of each storey should be kept below 3.2m.
4. Don't use bricks of crushing strength less than 35kg/cm^2 for storeyed building and of 50kg/cm^2 for 2-3 storeyed building. Only solid and sound bricks/ concrete blocks should be used.

5. Provide a R.C.C band of 4 inch thickness throughout the run along wall at lintel level passing over doors and windows.
6. The thickness of load bearing wall should be at least 200mm
7. The clear width between a door and nearest window should not be less than 600mm.
8. Location of a door or window from edge of a wall shall be 600mm minimum.

ROLE OF GEOTECHNICAL ENGINEERS

1. Gathering and analysing data.
2. Looking at the risk of geological hazards and making sure any factors affecting engineering works are identified and managed.
3. Advising on procedures required and the suitability of construction materials
4. Using specialist computer software to create analytical 2D and 3D models

5. Consulting geological maps and aerial photographs to advise on site selection.
6. Assisting with the design of built structures, using specialised computer software or calculations
7. Planning detailed field investigations by drilling and analysing samples of deposits or bedrock.
8. Supervising ground investigations and budgets
9. Advising and testing a range of construction materials including sand, gravel, bricks and clay

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10. Making recommendations on the proposed use of a site
 11. Managing staff, including other engineering geologists, geotechnical engineers, consultants and contractors
 12. Working to preserve and protect the physical environment
 13. Analysing sites and designs for environmentally sensitive developments, such as landfill.

Types of Foundation Structures

Foundations are mainly of two types:

- (i) shallow foundation
- (ii) deep foundations

Shallow Foundation

(A) Shallow foundations

Shallow foundations are used when the soil has sufficient strength within a short depth below the ground level. They need sufficient plan area to transfer the heavy loads to the base soil. These heavy loads are sustained by the reinforced concrete columns or walls (either of bricks or reinforced concrete) of much less areas of cross-section due to high strength of bricks or reinforced concrete when compared to that of soil.

The strength of the soil, expressed as the safe bearing capacity of the soil which is normally supplied by the geotechnical experts to the structural engineer. Shallow foundations are also designated as footings.

Deep foundation

If the depth of footing greater or equal to the Width of footing, it is known as the deep Foundation. Deep Foundation is used Where the bearing capacity of the soil is very low. The load coming from the superstructure is further transmitted vertically to the soil. There are Three Major Types of Deep Foundation, and Their uses in construction are discussed below.

Types of deep Foundation

1. Pile Foundation
2. Pier Foundation
3. Well Foundation

Economic role of transportation

Economics involves production, distribution and consumption of goods and services. People depend upon the natural resources to satisfy the needs of life but due to non uniform surface of earth and due to difference in local resources, there is a lot of difference in standard of living in different societies. So there is an immense requirement of transport of resources from one particular society to other.

Road Safety Signs

Its Primarily of Three Types:

1. Mandatory Signs
2. Cautionary Signs
3. Informatorily Signs

				
STOP	GIVE WAY	STRAIGHT PROHIBITOR NO ENTRY	PEDESTRIAN PROHIBITED	HORN PROHIBITED
				
NO PARKING	NO STOPPING OR STANDING	SPEED LIMITED	RIGHT HAND CURVE	LEFT HAND CURVE
				
RIGHT HAIR PIN BEND	LEFT HAIR PIN BEND	NARROW ROAD AHEAD	NARROW BRIDGE	PEDESTRIAN CROSSING
				
SCHOOL AHEAD	ROUND ABOUT	DANGEROUS DIP	HUMP OR ROUGH	BARRIER AHEAD

Mandatory signs

				
Right Hand Curve	Left Hand Curve	Right Hair Pin Bend	Left Hair Pin Bend	Right Reverse Bend
				
Left Reverse Bend	Steep Ascent	Steep Descent	Narrow Road Ahead	Road Wideness Ahead
				
Narrow Bridge	Slippery Road	Loose Gravel	Cycle Crossing	Pedestrian Crossing
				
School Ahead	Men at Work	Cattle	Falling Rocks	Ferry

Cautionary Signs



Informatorily Signs

Various road safety measures

1. As they say speed thrills and also kills. Keep a rein on your speed, so that one is in better control of the vehicle
2. Obey traffic rules. Traffic rules have been designed with safety foremost in mind. It takes into account safety for all, including pedestrians.
3. Wear seat belts. Seat belts are life saving. A statistic has revealed that 63% of those who died in accidents had not strapped on their seat belts.

4. Pedestrians and children come first – Preference is to be given to pedestrians and children crossing the road.
5. Read caution signs as Caution signs are very important as they warn about accident prone spots so that the driver can reduce the speed of the vehicle.
6. Adopt antiskid break systems in the cars, so that the risk of cars skidding can be reduced.
7. Air bags in vehicles are a must to cushion the impact in case of accidents.

Traffic calming

Traffic calming is applied to local streets with the aim of lowering traffic speeds and volumes, and sometimes, preventing particular types of vehicle travelling through an area. These aims are achieved by treating an area rather than a single section of road.

Traffic calming treatments cause drivers to change their driving pattern. Usually they have to reduce their speed but sometimes drivers are exposed to something undesirable (e.g. rumbling, or delays) which encourages them to choose a different route. There is a wide range of traffic calming treatments available. These include roundabouts, kerb build-outs, speed humps, raised tables, entry treatments, speed cushions, modified intersections and many others

WATER QUALITY STANDARDS

Sl No.	Characteristic	Requirement (Acceptable Limit)	Permissible Limit in the Absence of Alternate Source
(1)	(2)	(3)	(4)
i)	Colour, Hazen units, <i>Max</i>	5	15
ii)	Odour	Agreeable	Agreeable
iii)	pH value	6.5-8.5	No relaxation
iv)	Taste	Agreeable	Agreeable
v)	Turbidity, NTU, <i>Max</i>	1	5
vi)	Total dissolved solids, mg/l, <i>Max</i>	500	2 000

Sl No.	Characteristic	Requirement (Acceptable Limit)	Permissible Limit in the Absence of Alternate Source
(1)	(2)	(3)	(4)
i)	Aluminium (as Al), mg/l, <i>Max</i>	0.03	0.2
ii)	Ammonia (as total ammonia-N), mg/l, <i>Max</i>	0.5	No relaxation
iii)	Anionic detergents (as MBAS) mg/l, <i>Max</i>	0.2	1.0
iv)	Barium (as Ba), mg/l, <i>Max</i>	0.7	No relaxation
v)	Boron (as B), mg/l, <i>Max</i>	0.5	1.0
vi)	Calcium (as Ca), mg/l, <i>Max</i>	75	200
vii)	Chloramines (as Cl ₂), mg/l, <i>Max</i>	4.0	No relaxation
viii)	Chloride (as Cl), mg/l, <i>Max</i>	250	1 000
ix)	Copper (as Cu), mg/l, <i>Max</i>	0.05	1.5
x)	Fluoride (as F) mg/l, <i>Max</i>	1.0	1.5
xi)	Free residual chlorine, mg/l, <i>Min</i>	0.2	1

xii)	Iron (as Fe), mg/l, <i>Max</i>	0.3	No relaxation
xiii)	Magnesium (as Mg), mg/l, <i>Max</i>	30	100
xiv)	Manganese (as Mn), mg/l, <i>Max</i>	0.1	0.3
xv)	Mineral oil, mg/l, <i>Max</i>	0.5	No relaxation
xvi)	Nitrate (as NO ₃), mg/l, <i>Max</i>	45	No relaxation
xvii)	Phenolic compounds (as C ₆ H ₅ OH), mg/l, <i>Max</i>	0.001	0.002
xviii)	Selenium (as Se), mg/l, <i>Max</i>	0.01	No relaxation
xix)	Silver (as Ag), mg/l, <i>Max</i>	0.1	No relaxation
xx)	Sulphate (as SO ₄) mg/l, <i>Max</i>	200	400
xxi)	Sulphide (as H ₂ S), mg/l, <i>Max</i>	0.05	No relaxation
xxii)	Total alkalinity as calcium carbonate, mg/l, <i>Max</i>	200	600
xxiii)	Total hardness (as CaCO ₃), mg/l, <i>Max</i>	200	600
xxiv)	Zinc (as Zn), mg/l, <i>Max</i>	5	15

Rain water Harvesting

Rainwater harvesting (RWH) is a simple method by which rainfall is collected for future usage. The collected rainwater may be stored, utilized in different ways or directly used for recharge purposes. With depleting groundwater levels and fluctuating climate conditions, RWH can go a long way to help mitigate these effects. Capturing the rainwater can help recharge local aquifers, reduce urban flooding and most importantly ensure water availability in water-scarce zones.

Environmental Acts and Regulation

The Environment (Protection) Act, 1986

The Environment (Protection) Act, 1986 authorizes the central government to protect and improve environmental quality, control and reduce pollution from all sources, and prohibit or restrict the setting and /or operation of any industrial facility on environmental grounds.